

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Biotechnology) (Credit System) Degree Examinations

April 2023

21BT302 - BIOCHEMISTRY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Define isomerism. With examples describe the isomerism in carbohydrates.	10	L*2	1	1
b)	What are glycoconjugates? Distinguish among the different types of glycoconjugates.	10	L2	1	1
2. a)	Justify the wide biological importance of lipids.	10	L2	1	1,2
b)	Differentiate between homo and hetero polysaccharides. Highlight the biological importance of these polysaccharides.	10	L2	1	1,2
3. a)	Justify carbohydrates as informational molecules.	10	L2	1	1,2
b)	Explain the biological roles played by sterols and steroid hormones.	10	L2	1	1,2
	Unit – II				
4. a)	With a neat labelled diagram, explain the secondary structure of DNA.	10	L2	2	1
b)	Appraise mechanisms in formation of protein secondary structures.	10	L2	2	1
5. a)	Classify the amino acids based on the functional groups.	10	L2	2	1
b)	Describe the secondary and tertiary structure of RNA.	10	L2	2	1
6. a)	Differentiate fibrous and globular proteins with suitable examples.	10	L2	2	1
b)	Draw the structures of a DNA nucleotide and explain the 3 different bonds in stabilizing the DNA primary structure.	10	L2	2	1
	Unit – III				
7. a)	Elucidate the mechanism of protein degradation and formation of urea.	10	L2	3	1
b)	Outline the process of Gluconeogenesis.	10	L2	3	1
8. a)	Explain the processes involved in the degradation of purines and pyrimidines.	10	L2	3	1
b)	Describe the beta oxidation of fatty acids.	10	L2	3	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
